Determinants of Household Participation in Securities Markets

An Empirical Analysis of the 2025 SEBI Household Survey

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The Problem

Only 25.5% of Indian households invest in securities.

Despite the post-2020 retail boom and zero-commission platforms, participation remains sharply stratified by socioeconomic position, geography, and information access.

Four research questions:

1. Who is likely to participate?
2. What keeps the 74.5% out?
3. Which securities do investors prefer, and for how long?
4. Can we predict these outcomes from basic profiles?

Sample

Total respondents	109,430
Active investors	27,882 (25.5%)
Non-investors	81,548 (74.5%)
Geographic zones	4
States / UTs covered	35

Methods

Logistic regression (HC3 robust SE, survey-weighted), K-means clustering, Histogram-based Gradient Boosting

Dataset and Schema: Details

Source: 2025 SEBI Household Survey, nationally representative, pre-computed survey weights.

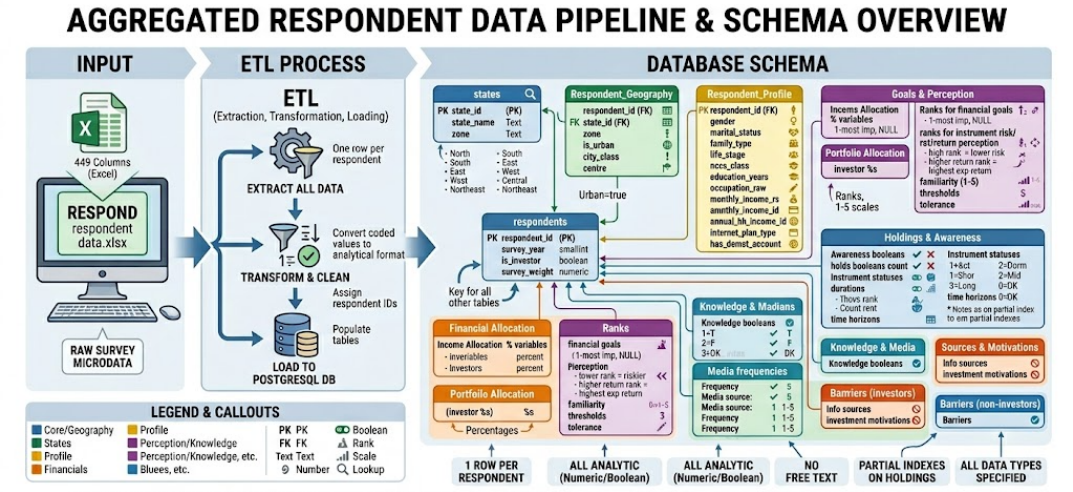
ETL pipeline: 449 raw columns transformed into boolean, integer, and ordinal types; loaded into a normalized PostgreSQL schema (17 domain-specific tables anchored by `respondent_id`).

Conditional routing:

- Investors: sub-module SS_B3 (info sources, motivations), $n = 9{,}797$
- Non-investors: sub-module SS_BB2 (barriers), $n = 14{,}794$

All group percentages computed against the routing-eligible population.

Dataset and Schema: Architecture



Methodology (I): Econometric Model

Multivariate Participation Logic

$$\begin{aligned}\text{logit}(p_i) = \beta_0 &+ \beta_1 \text{Awareness}_i + \beta_2 \ln(\text{Income}_i) + \beta_3 \text{Education}_i \\ &+ \beta_4 \text{Male}_i + \beta_5 \text{Urban}_i + \beta_6 \text{Risk}_i \\ &+ \sum_j \delta_j \text{Cohort}_{ij} + \sum_l \lambda_l \text{Zone}_{il} + \epsilon_i\end{aligned}$$

- **Dependent Variable:** p_i is the probability of market participation ($Y = 1$).
- **Survey Weighting:** All coefficients estimated using inverse probability weighting to match national demographics.
- **Robustness:** HC3 Huber-White robust standard errors to account for complex survey design and non-normality.

Methodology (II): Unsupervised & Predictive Layer

Barrier Archetypes (Finding 2)

- **Algorithm:** K-Means clustering.
- **Feature Space:** 18-dimensional binary vector of self-reported barriers.
- **Optimization:** Elbow method and Silhouette scores for cluster validation ($n = 3$ archetypes).

Predictive Modeling (Layer)

- **Model:** Histogram-based Gradient Boosting.
- **Capability:** Handles categorical features natively; captures complex non-linear interactions.
- **Validation:** 5-fold cross-validation; AUC-ROC as primary metric.

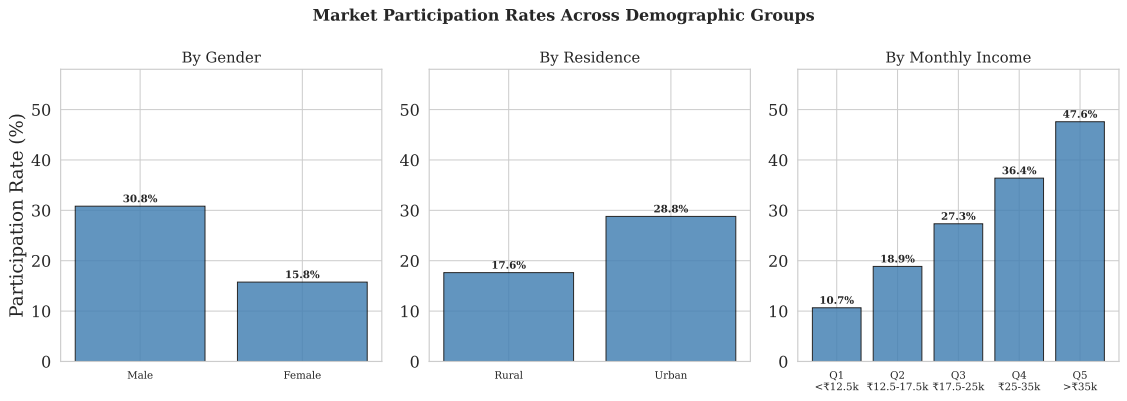
EDA: Who Participates?

Steepest univariate signals:

- Income gradient: 10.7% (Q1, <Rs. 12,500/mo) to 47.6% (Q5, >Rs. 35,000/mo): a **4.5**× spread
- Gender gap: 30.8% (male) vs 15.8% (female), persists across all income levels
- Urban premium: 28.8% vs 17.6% (rural)
- Millennials (28–43) lead cohorts at 28.1%, exceeding both Gen Z (24.3%) and Gen X (22.9%)

These univariate signals motivate the multivariate regression: which factors survive joint controls?

EDA: Participation Demographics



EDA: Awareness Gap and Holdings

Awareness gradient (all 109,430 respondents):

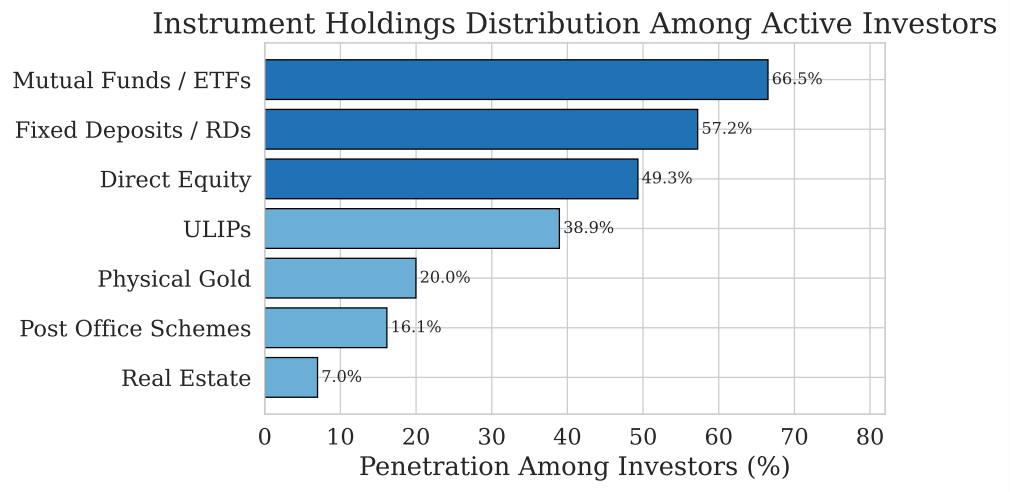
- Fixed Deposits: **99.1%**
- Mutual Funds / ETFs: 67.0%
- Direct Equity: 62.0%
- Derivatives: **17.9%**

Holdings among investors:

- MF/ETF: 66.5% FD/RD: 57.2%
- Direct equity: 49.3%

MFs and FDs *co-dominate*: Indian retail investors layer market-linked products on top of traditional savings rather than replacing them.

EDA: Instrument Awareness and Holdings



Finding 1: What Drives Market Entry?

Product awareness is the single strongest predictor (OR = 2.22), exceeding:

Income (log)	OR = 1.60
Education	OR = 1.56
Male gender	OR = 1.53
Urban	OR = 1.45

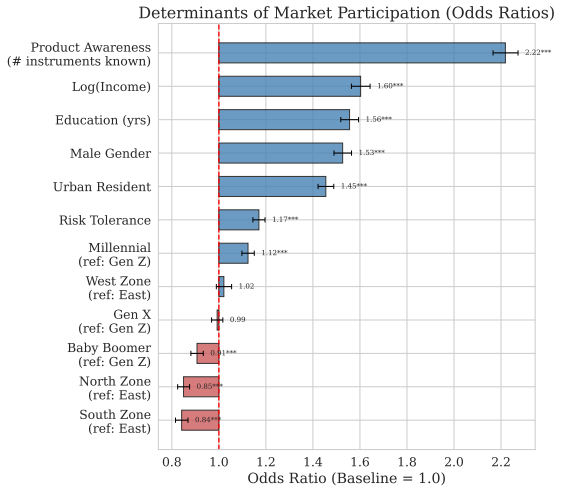
Partial mediation: income and education partly work *through* awareness, but awareness retains a large independent effect after SES controls.

Two predictors return null results (Gen X, West zone): the model does not produce spurious significance at $N \approx 100{,}000$.

Policy implication

Awareness is a **distinct, actionable lever**, not merely an SES proxy.

Finding 1: Odds Ratios for Participation



Finding 2: Three Archetypes of Non-Participation

K-means on 18 self-reported barriers identifies three profiles:

Trust-Deficient (46.3%)

Persistent across income levels and urban cohorts. Requires institutional accountability: grievance redressal, IPF promotion, broker transparency.

Knowledge-Gated (29.0%)

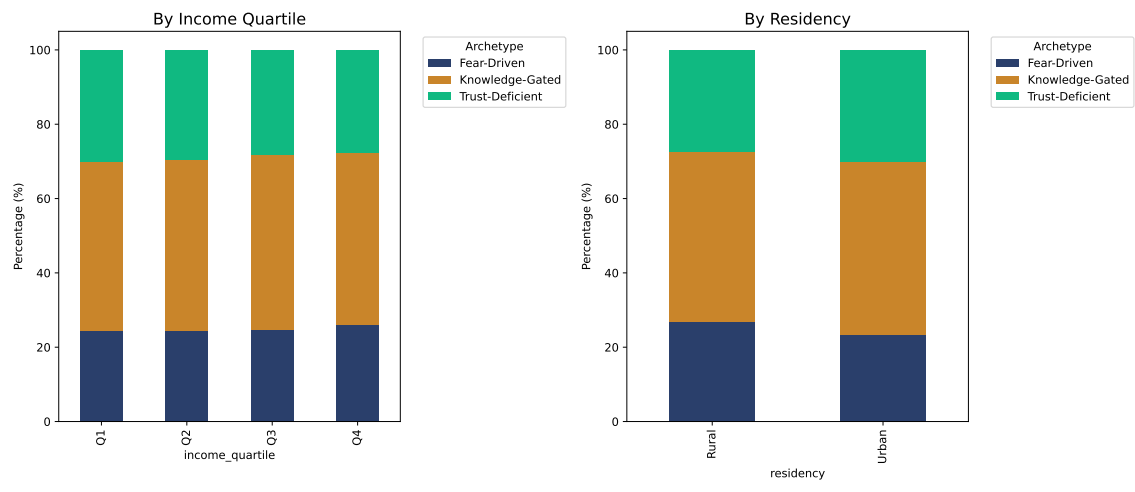
Concentrated in rural areas and lowest income quartile. Needs geographically targeted vernacular literacy delivery.

Fear-Driven (24.8%)

Responds to loss-framing education and first-loss protection onramps (SGB, RBI Floating Rate Bonds).

A single awareness campaign cannot address all three.

Finding 2: Archetype Demographics



Finding 3: Behavioral Traps Inside the Market

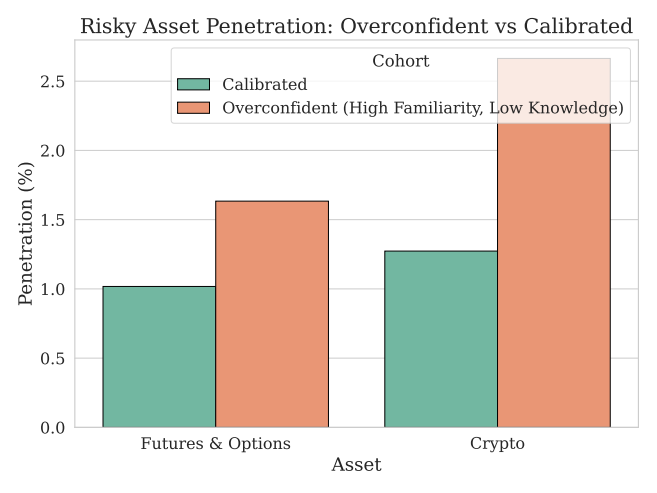
Dunning-Kruger effect: Investors with high perceived familiarity but low objective knowledge hold F&O and crypto at significantly higher rates than calibrated peers.

Finfluencer pipeline: 5.0% of investors cite social media as primary information source; this cohort is disproportionately quick-gains oriented relative to professionally-advised investors.

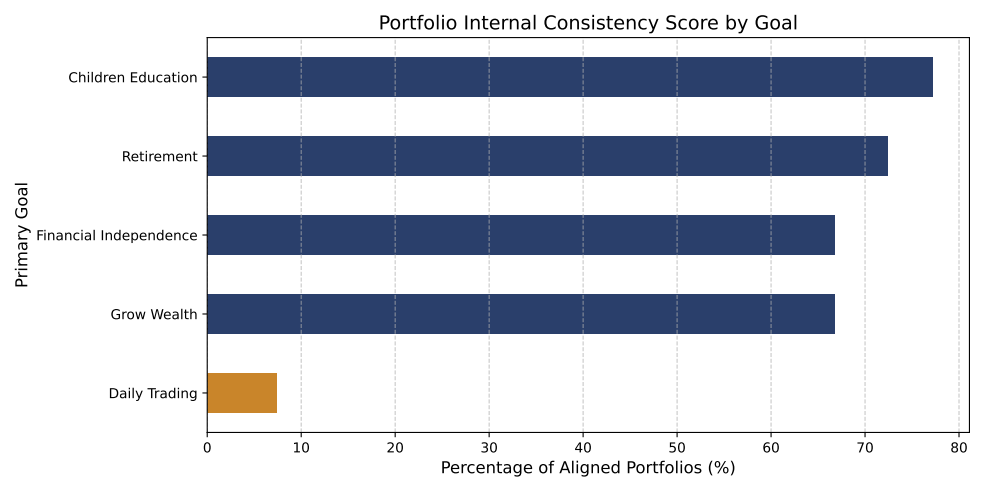
Goal-portfolio gap: The "Daily Trading" cohort has a goal-portfolio consistency score of 7.3%: they identify as speculators without holding the instruments that operationalize that strategy.

These phenomena compound: aspirational investors, shaped by finfluencer content, are the segment most exposed to adverse outcomes.

Finding 3: Dunning-Kruger and Holdings



Finding 3: Goal-Portfolio Consistency



Finding 4: Investor Education Programme Impact

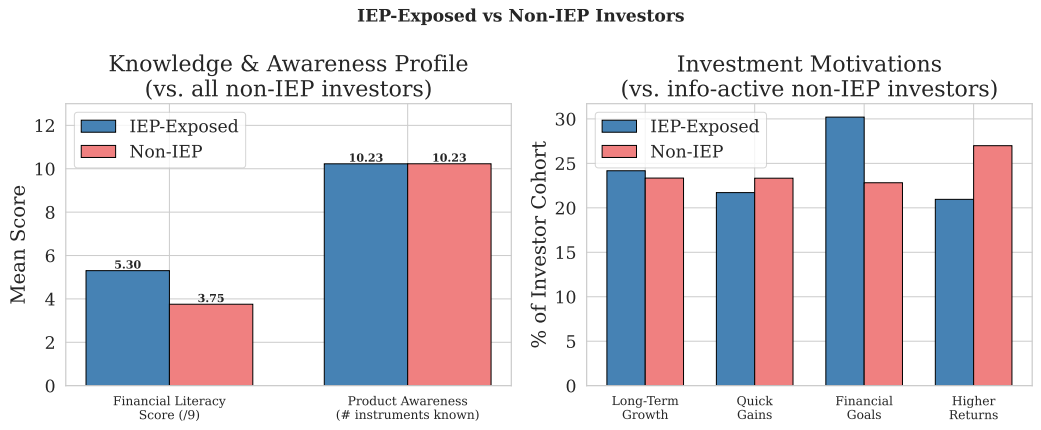
2,768 investors (9.9%) cite SEBI or industry IEP as a top information source.

Metric	IEP	Non-IEP
Knowledge score	**5.30**	3.75
Products aware	10.23	10.23
Goal-based (%)	**30.2**	22.8
Quick gains (%)	21.7	23.3
Holds FDs (%)	49.3	**55.1**

The 41% knowledge advantage is specific to *conceptual depth*, not product breadth (awareness is near-identical).

IEP functions as both a knowledge-deepening and a philosophical-reorientation intervention. Current reach: 9.9%.

Finding 4: IEP Analysis



Predictive Layer: Machine Learning Results

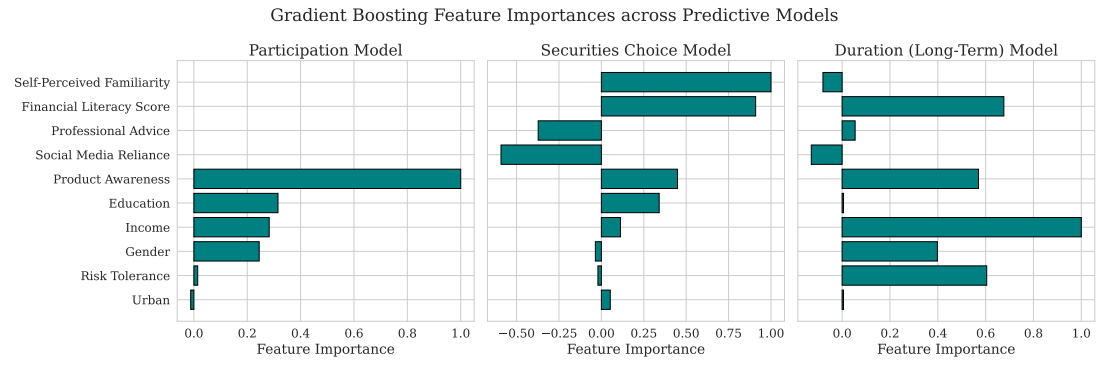
Model	Task	AUC
M1	Participation	**0.814**
M2	Asset choice	**0.820**
M3	Horizon	0.612

Feature importance shift:
At the entry layer, *product awareness* dominates. Inside the market, *perceived familiarity* and *objective knowledge* take over.

Investment horizon (M3) is the hardest prediction, consistent with it being driven by situational factors outside a cross-sectional feature space.

All three models include `n_products_aware` alongside demographic and behavioral features.

Predictive Layer: Feature Importance



**Note: Negative permutation importance indicates high-variance predictors (e.g., social media usage) that may appear noisy when shuffled individually. However, HistGradientBoosting utilizes these features for critical conditional interactions; their removal breaks high-precision decision branches, leading to lower aggregate AUC.*

Policy Recommendations

A. Prioritize product awareness
OR = 2.22; campaigns should name specific instruments (MF, ETF, SIP) for North and South zones.

B. Segment non-investor interventions
Trust-Deficient: regulatory transparency. Knowledge-Gated: vernacular rural delivery. Fear-Driven: first-loss protection onramps.

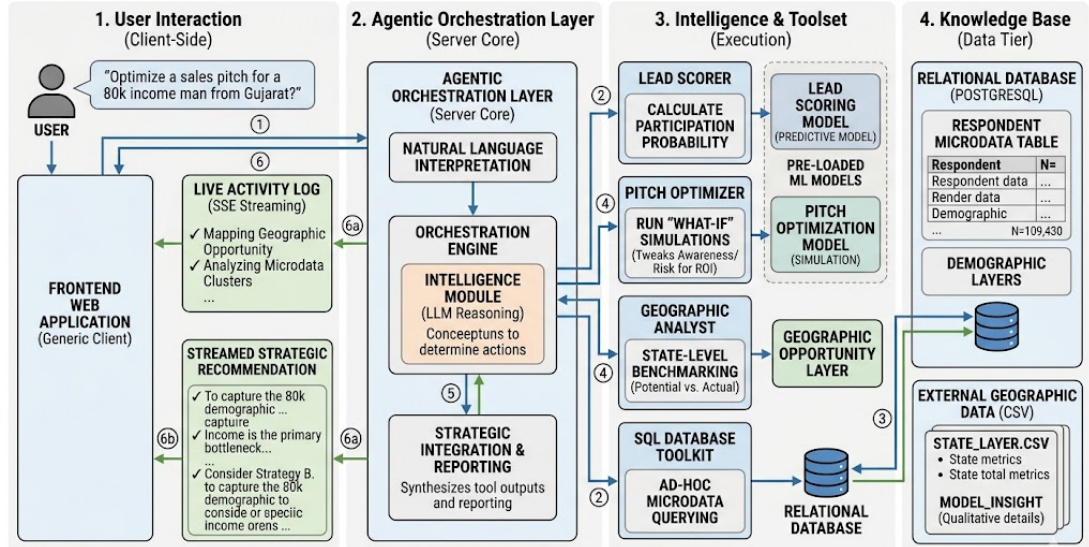
C. Scale IEP reach
From 9.9% to employer mandates and Jan Dhan partnerships. Content should emphasize conceptual depth, not procedural literacy (KYC awareness is already high). Goal-aligned investors show 72–77% portfolio consistency, confirming goal-based framing works.

D. Close the gender gap
OR = 1.53 for male gender after full controls. SHG-linked investment collectives, women-targeted IEP content, lower minimum investment thresholds.

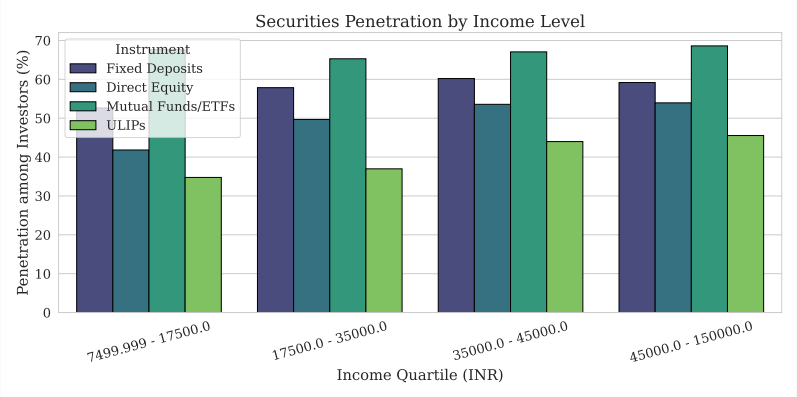
E. Regulate finfluencers
5% usage associated with quick-gains orientation. Mandatory qualification disclosure, prohibited performance claims, platform liability for unregistered advice.

F. Gate derivatives on objective literacy
Overconfident investors and aspirational daily traders (7.3% consistency) reach leveraged instruments without adequate knowledge. Replace self-certification with a brief objective literacy assessment.

Implementation: Stockbroker Sales AI

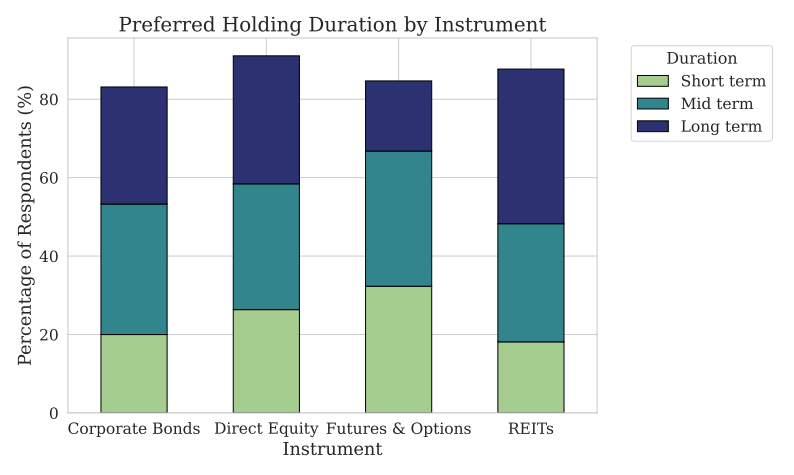


Appendix: Holdings by Income Quartile



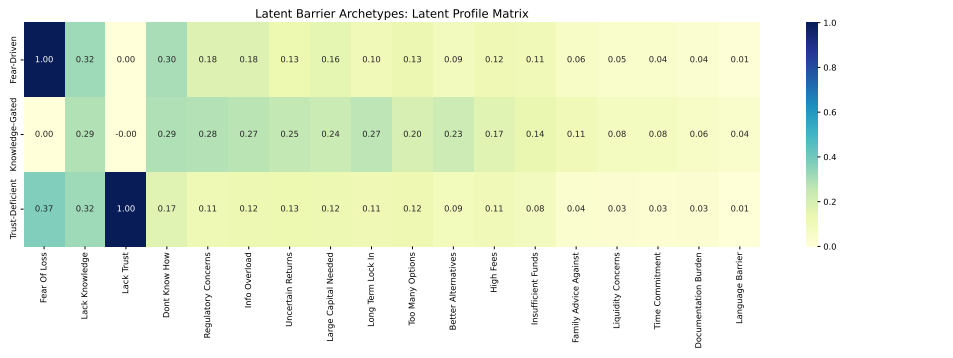
Market-linked instruments scale sharply with wealth.

Appendix: Investment Horizons by Asset Class



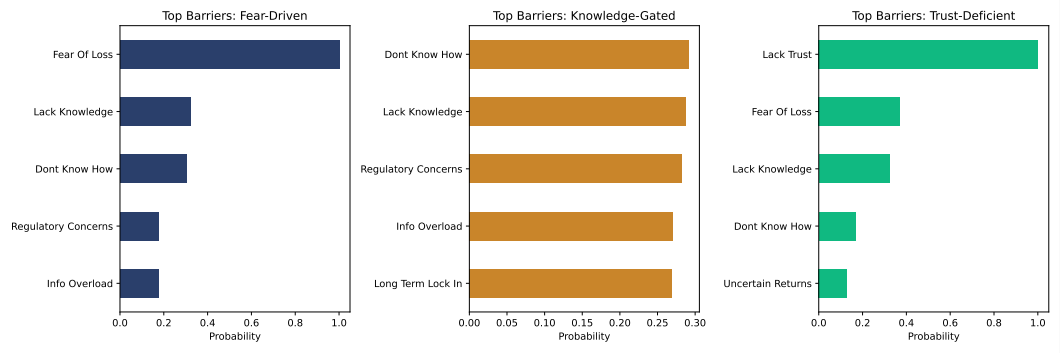
Equities and MFs held long-term; derivatives skew short-term.

Appendix: Barrier Archetype Profiles (Centroids)



Centroid matrix of the 18-dimensional barrier space. Values represent the probability of each barrier being present in the archetype profile.

Appendix: Archetype Defining Peaks



Top 5 barriers for each archetype. These peaks define the strategic naming: Fear-Driven (Fear of Loss), Knowledge-Gated (Lack Knowledge), and Trust-Deficient (Lack Trust).

Appendix: Full Logistic Regression Table

Predictor	**OR**	**95% CI**	z	Sig.
Product Awareness	2.218	[2.166, 2.272]	65.2	***
Log(Income)	1.603	[1.564, 1.643]	37.4	***
Education (yrs)	1.556	[1.518, 1.594]	35.6	***
Male	1.526	[1.490, 1.564]	33.9	***
Urban	1.455	[1.422, 1.489]	32.0	***
Risk Tolerance	1.170	[1.145, 1.196]	14.0	***
Millennial	1.124	[1.098, 1.151]	9.8	***
Gen X	0.993	[0.969, 1.017]	−0.6	
Baby Boomer	0.907	[0.881, 0.934]	−6.6	***
North	0.850	[0.825, 0.876]	−10.7	***
South	0.842	[0.816, 0.869]	−10.7	***
West	1.021	[0.989, 1.054]	1.3	

$N = 97{,}626$ McFadden $R^2 = 0.290$ *** $p < 0.001$, HC3 robust SE